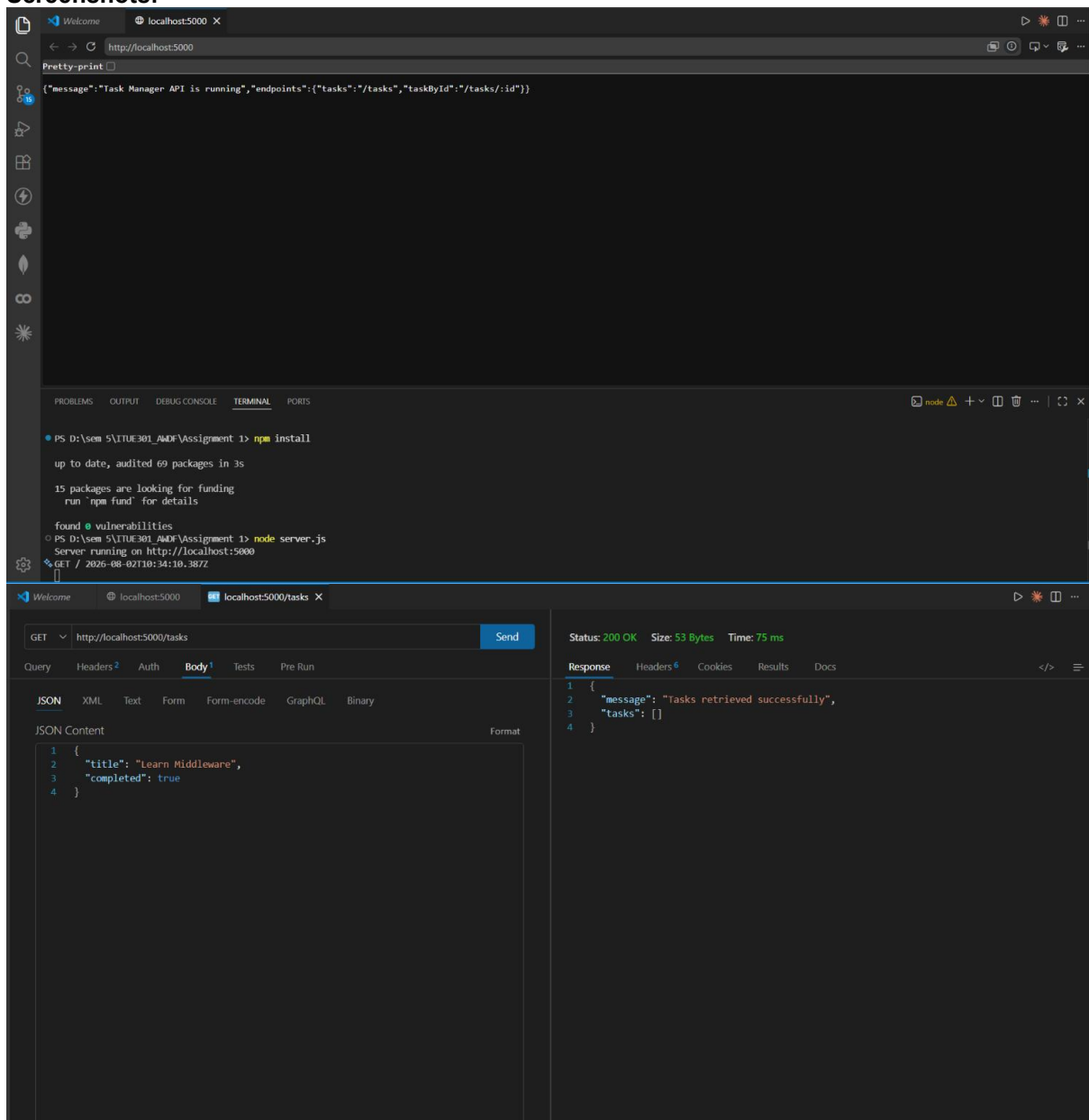


Subject: Advanced Web Development Frameworks (ITUE301) | Semester: 5th

Practical Statement:

Evaluate the Task Management REST API developed in Practical 4 using the Richardson Maturity Model (Levels 0–3). Identify the maturity level achieved by the API, justify the evaluation based on REST principles, improve the API if required to satisfy at least Level 2, and document the complete evaluation, HATEOAS awareness example, and observations in the MATURITY.md file.

Screenshots:



The screenshot displays a Visual Studio Code editor with two main panels. The top panel shows a REST client interface with a 'Send' button and a 'Body' tab selected. The body contains a JSON object:

```
{ "title": "Learn Middleware", "completed": true }
```

. The bottom panel shows a terminal window with the following output:

```
PS D:\sem 5\ITUE301_AmdF\Assignment 1> npm install
up to date, audited 69 packages in 3s
15 packages are looking for funding
run 'npm fund' for details
found 0 vulnerabilities
PS D:\sem 5\ITUE301_AmdF\Assignment 1> node server.js
Server running on http://localhost:5000
GET / 2026-08-02T10:34:10.387Z
```

Student ID: 24IT087

The image displays three sequential screenshots of a REST client interface, showing the process of creating and updating a task.

Top Screenshot (GET Request):

- Method:** GET
- URL:** http://localhost:5000/tasks/1
- Status:** 404 Not Found
- Size:** 26 Bytes
- Time:** 29 ms
- Response Body:**

```
{  "error": "Task not found"}
```

Middle Screenshot (POST Request):

- Method:** POST
- URL:** http://localhost:5000/tasks
- Status:** 201 Created
- Size:** 94 Bytes
- Time:** 24 ms
- Request Body:**

```
{  "title": "Learn REST",  "completed": false}
```
- Response Body:**

```
{  "message": "Task created successfully",  "task": {    "id": 1,    "title": "Learn REST",    "completed": false  }}
```

Bottom Screenshot (PUT Request):

- Method:** PUT
- URL:** http://localhost:5000/tasks/1
- Status:** 200 OK
- Size:** 94 Bytes
- Time:** 24 ms
- Request Body:**

```
{  "title": "Learn REST",  "completed": false}
```
- Response Body:**

```
{  "message": "Task updated successfully",  "task": {    "id": 1,    "title": "Learn REST",    "completed": false  }}
```

The screenshot displays a REST client interface at the top and a code editor at the bottom. The REST client shows a DELETE request to `http://localhost:5000/tasks/1`. The code editor shows the `MATURITY.md` file with the following content:

```

11 # Introduction
12 THE RACHARSON MATURITY MODEL IS USED TO EVALUATE HOW RESILIENT AN API IS.
13
14 It consists of four maturity levels.
15
16 - Level 0 - Single endpoint
17 - Level 1 - Resource-based URLs
18 - Level 2 - HTTP methods and Status Codes
19 - Level 3 - HATEOAS
20
21 The Task Management API was evaluated against all four levels.
22
23 ---
24
25 # Evaluation Table
26
27 | Level | Criterion | Does API satisfy? | Evidence |
28 |-----|-----|-----|-----|
29 | Level 0 | Single endpoint architecture | ❌ No | API uses multiple endpoints instead of one endpoint |
30 | Level 1 | Separate resources | ✅ Yes | '/tasks' and '/tasks/:id' are separate resources |
31 | Level 2 | Proper HTTP methods | ✅ Yes | GET, POST, PUT and DELETE methods are used |
32 | Level 2 | Proper HTTP status codes | ✅ Yes | 200 OK, 201 Created, 404 Not Found are returned |
33 | Level 3 | HATEOAS | ❌ No | Hypermedia links are not included in responses |
34
35 ---
36
37 # Detailed Evaluation
38
39 ## Level 0
40
41 Level 0 APIs expose only a single endpoint and all operations are performed through that endpoint.
42
43

```

Github Repo Link:

https://github.com/digishasavaliya/ITUE301_AWDF.git